State of Wisconsin
Department of Natural Resources
Bureau of Drinking Water and Groundwater, DG/5
PO Box 7921, Madison WI 53707-7921
dnr.wi.gov

Water Main Submittal Checklist

Form 3300-066 (R 02/2024)

Notice: This form is authorized by s. 281.41, Wis. Stats., and ss. NR 108.04 (2)(a), 811.08 (1), and 811.09(1)(h)3., Wis. Adm. Code. Completion of this form or a similar form approved by the Department of Natural Resources (DNR) is mandatory. Personal information collected will be used for administrative purposes and may be provided to requesters to the extent required by Wisconsin's Public Records Law [ss. 19.31-19.39, Wis. Stats.]. Unless otherwise noted all citations refer to Wis. Adm. Code.

WATER MAIN SUBMITTAL INSTRUCTIONS:

The following is a listing of information that must be submitted when requesting an approval of a water main project:

1. A completed Water System Approval Request (DNR Form No. 3300-260) has been included? (●) Yes ( ) No
2. One set of plans sealed by a Professional Engineer has been included in conjunction with a water main only submittal? (●) Yes ( ) No ( ) N/A
3. Three sets of plans sealed by a Professional Engineer have been included in conjunction with a facilities submittal? ( ) Yes ( ) No (●) N/A
4. The water main plans and specifications provide both a plan and profile of the water main. The plans show the location of the water main within street rights of way or easements and the location of other utilities, appurtenances in the plan and profiles. [NR 811.09(1)(a) & (h)] (●) Yes ( ) No
5. One set of specifications sealed by a Professional Engineer has been included in conjunction with a water main only submittal? ( ) Yes ( ) No (●) N/A

   The Specifications Section of the checklist must be completed if specifications are submitted. The submittal of specifications is not necessary if the community already has specifications on file with the Department of Natural Resources (that are not older than five years) or is using the latest edition of the Standard Specifications for Sewer and Water Construction in Wisconsin.
6. Three sets of specifications sealed by a Professional Engineer have been included in conjunction with a facilities submittal? ( ) Yes ( ) No (●) N/A
7. A plan or plans has been included that shows the location of the proposed project in relation to the rest of the distribution system? (●) Yes ( ) No ( ) N/A
8. An approval letter from the owner of the distribution system has been included, unless the applicant is employed or retained by the waterworks owner? (●) Yes ( ) No ( ) N/A

All sections of this checklist must be completed for every submittal; excepting, that if specifications are on file for the municipality or standard specifications are to be used, the section under specifications may be omitted. If it is felt that a question on the checklist does not apply to a particular project, indicate this with N/A and explain the reason.

A. General Information

Name of Municipality/Sanitary District/Other	Name/Number of Project
Village of Pleasant Prairie	Zilbur Industrial 3 at Lakeview

Type of Project: ( ) Relay (●) Extension

Is this proposed water main extension located within the boundaries of the community providing the source of water? (●) Yes ( ) No

Is an interconnection between two water distribution systems proposed? ( ) Yes (●) No

B. Specifications

Water mains will be constructed in accordance with: (select one of the following three options)

(●) 1. Standard Specifications for Sewer and Water Construction in Wisconsin (6TH Edition)

( ) 2. Standard specifications for municipality already on file with DNR: Approval number for specifications _____

Date of Approval _____ Are specifications on file with DNR less than 5 years old? ( ) Yes ( ) No

( ) 3. Specifications submitted with the plans (fill out the following section)

C. Design Specifics

1. Minimum horizontal (center to center) separation distance between water main and existing or future sanitary sewer 8 ft., storm sewer 8 ft. [NR811.74(2)] Minimum horizontal separation distance between fire hydrant drains and sanitary sewers, storm sewers, or storm sewer inlets 8 ft. [NR811.71(4)]
2. What is the minimum depth of water main cover throughout the project? [NR811.73(2)(e)] 6 ft.,
3. Where water mains cross over sewers, the minimum vertical separation distance (edge to edge) is 6 inches. Where water mains cross under sewers, the minimum vertical separation distance (edge to edge) is 18 inches. [NR811.74(3)]
4. Will a common trench be used in any portion of the project? [NR811.74(2)(b)] ( ) Yes (●) No
5. Will the following minimum horizontal separation distances be maintained between the water main and the contamination sources listed?[NR811.75]
   (A) Three feet between a water main and a stormwater collection system treatment unit, such as a bioswale or permeable paver system? ( ) Yes ( ) No (●) N/A

C. Design Specifics (cont'd)

(B) Eight feet between a water main and a POWTS holding, treatment, or dispersal component, sanitary sewer lift-station, grave site, stormwater detention pond, retention pond, or infiltration basin? — ◉ Yes ○ No ○ N/A

(C) Twenty-five feet to a buried fuel tank or main? — ○ Yes ○ No ◉ N/A

(D) Fifty feet to a sanitary landfill or agricultural, industrial, commercial, or municipal wastewater treatment plant treatment unit, lagoon, or storage structure? — ○ Yes ○ No ◉ N/A

6. Does the municipality have an erosion control ordinance? [NR811.09(2)] — ◉ Yes ○ No

(A) If yes, will compliance with the ordinance be required for this project? — ◉ Yes ○ No ○ N/A

(B) Do the plan sheets show the erosion control provisions? — ◉ Yes ○ No

(C) Do the specifications require that the erosion control measures be in place before construction begins and be maintained during construction? — ◉ Yes ○ No

(D) Do the required erosion control provisions comply with the technical standards of ch. NR151? — ◉ Yes ○ No

7. Are valves provided at each intersection and at intermediate points so spacing does not exceed 800 feet? [NR811.70(9)] — ◉ Yes ○ No

8. Are hydrants provided at each intersection and at intermediate points so spacing does not exceed 600 feet? [NR811.71(1)] — ◉ Yes ○ No

9. Are hydrants or other flushing devices capable of flow velocities of at least 3 feet per second in the water main installed downstream of the last service at all dead-ends? [NR811.71(7)] — ◉ Yes ○ No

10. Will any watermain stubs 20 feet or greater in length be installed? [NR811.70(8)] — ○ Yes ◉ No

11. If groundwater may rise above hydrant drain ports, will the drain ports be plugged and operational procedures established for pumping the hydrant barrels dry during freezing weather? [NR811.71(4)] — ○ Yes ○ No ◉ N/A

12. Is there a history of external corrosion problems with buried pipe in the project area? [NR811.69(4)] — ○ Yes ◉ No

13. Do the proposed water mains pass through or adjacent to a landfill or chemical spill area that may adversely impact the piping material or gaskets? [NR811.69(5)] — ○ Yes ◉ No

14. Do the proposed water mains pass through a wetland area? [NR811.70(3)] — ◉ Yes ○ No

If yes:

(A) Is the extent of the wetlands shown on the plans? — ◉ Yes ○ No

(B) Is a copy of the Army Corps of Engineers 404 permit included? — ○ Yes ◉ No

If no, explain:
The watermain will be directionally drilled under the wetland. The wetland will not be disturbed as part of the installation of the watermain.

Be aware it is Department policy to discourage construction through wetlands if practical alternatives exist. To satisfy the goals of this policy, the Department requests that you attach a justification for construction through a wetland that also includes discussion of other alternatives.

(C) Is justification for the construction through a wetland attached? — ○ Yes ◉ No

If no, explain:
The watermain will be directionally drilled under the wetland. The wetland will not be disturbed as part of the installation of the watermain.

(D) Is a copy of the Department's letter of Wetland Water Quality Certification of the 404 permit included? — ○ Yes ◉ No

If no, explain:
The watermain will be directionally drilled under the wetland. The wetland will not be disturbed as part of the installation of the watermain.

15. Do the proposed water mains pass through a floodway or floodplain? [NR811.70(2)]? — ○ Yes ◉ No

16. Does installation of the proposed water mains involve construction within 500 feet of the ordinary high water mark or over or under or in waters of the state? — ○ Yes ◉ No

17. Do the proposed water mains extensions include water main(s) that would divert water from the Great Lakes - St. Lawrence River Basin to the Mississippi River Basin? — ○ Yes ◉ No

18. If plans are submitted by someone other than the waterworks owner or authorized representative, is written acceptance of the waterworks owner included? [NR811.10] — ◉ Yes ○ No ○ N/A

C. Design Specifics (cont'd)

19. Do the proposed water mains involve construction of manholes, vaults, or other below grade structures containing shutoff valves, air relief valves, pressure reducing valves, or water meters? [NR 811.72] ○ Yes ◉ No

20. Do the proposed water mains involve any surface water crossings over 15 feet? [NR811.76] ○ Yes ◉ No

21. Do the proposed water mains involve any common casing crossings? [NR811.77] ○ Yes ◉ No

22. Will private mains or services be connected at more than one location, creating a loop back into the public system [NR811.68(3)]? ○ Yes ◉ No

23. Water main pipe material(s) to be used: PVC

Type of joint(s): Integral Elastometric Bell and Spigot Joints

Class and/or DR: C-900 [NR811.69]

24. If ductile iron pipe will be used, will it be enclosed in polyethylene wrap? ○ Yes ○ No ◉ N/A

25. Will proposed water mains serve existing structures having private wells? [NR810.15, NR810.16] ○ Yes ◉ No

26. Will installation of the water main(s) include dewatering well construction having a total capacity exceeding 70 gallons per minute? [NR812.09(4)(a)] ○ Yes ◉ No

27. On-site inspection of the proposed water main construction will be provided by (check all that apply): [NR811.11]

☐ Engineering firm ☒ Owner ☐ Other (specify) _____

28. Normal static pressures throughout the area to be served will range from _____ to _____ PSI [NR 811.66(1)(b) & NR 811.70(4)]

29. The area to be served is (check all that apply): ☐ Residential ☐ Commercial ☐ Industrial [NR 811.70(9) & NR 811.71(5)]

30. Calculate the minimum fire flow at any proposed hydrant. [NR811.70(5)&(6), NR811.71(3)]

Fire Flow Test

Location of residual hydrant: Intersection of 108th St and STH 31, northwest Corner

Location of flowing hydrant(s): 445 ft to the west of the residual hydrant

Distance between residual and flowing hydrant(s): 445 ft.

Static pressure at residual hydrant: 50.00 PSI Elevation of residual hydrant: 724 ft.

Flow test results: 2190 GPM at a residual pressure of 41 PSI

Conversion of flow test to 500 GPM equivalent yields 500 GPM @ 53.1 PSI

Calculations

Location of critical hydrant: Approximately 1,100 ft south of the Intersection of 108th and STH 31

Distance between critical and residual hydrants: 1097 ft.

Roughness coefficient ("C" factor): 120 Elevation of critical hydrant: 730.9 ft.

Head loss due to friction: 0.3 PSI Head loss/gain due to elevation: 3.0 PSI Total head loss: 3.8 PSI

Calculated available fire flow at critical hydrant is 500 GPM at 46.2 PSI

(Attach additional sheets if necessary.)

For computer generated models, output must include all losses, assumed flows, roughness coefficient, pipe lengths, pipe diameters, and a node map.

31. Water mains proposed:

Diameter[1] (inches)	Length[1] (feet)	Street name and/or easement description [NR811.67]
16	1,088	Along STH 31
6	7	Hydrant Stubs

[1]**Note:** Rule requirements of other state agencies pertinent to water mains, such as Wis. Adm. Code ch. PSC 184, should be reviewed to obtain all necessary approvals.

D. Additional Comments

Further comments on any previous items (use additional sheets if necessary):

E. Certification

I certify that I have examined the above information and found it to be correct, true and complete.

☐ This form may be signed electronically, pursuant to Wis. Stat. Chapter 137. By checking the electronic signature acknowledgement box and typing your name, you are expressing intent to sign this form and attesting that all information provided is true and accurate to the best of your knowledge. If you decline to sign this form electronically, you must physically sign this form and all other forms required for your project.

Printed Name of Professional Engineer		Wis. P.E. Number
Andrew Shoaf		42919-6
Email address	**Phone Number (incl. area code)**	**Fax Number (inc. area code)**
andrew.shoaf@pinnacle-engr.com	(262) 754-8888	
Signature of Professional Engineer		**Date Signed**
		10/9/2024

Flow Test Form

Project #	N/A
Project Name	N/A
Project Address	10790 Green Bay Rd.
City	Pleasant Prairie
State	WI
Date of Test	October 7, 2026
Time of Test	10 AM
Witnessed by	J.F. Ahern
Performed by	Village of Pleasant Prairie

Static Pressure	50.00	PSI
Residual Pressure	41.00	PSI
Pitot pressure (p)	26.00	
Butt/Orifice opening size (d)	4.00	
Hydrant co-efficient (c)	0.80	

FM Global projects—a minimum co-efficient of 0.80 is to be used to determine GPM flowing.

Formula
NFPA 291 Section 4.7.3

$Q = 29.84cd^2\sqrt{P}$

where:
c = coefficient of discharge (Figure 4.7.1)
d = diameter of the outlet (in.)
p = pitot pressure (velocity head) (psi)

GPM Results	2,190.00
Hydrant Numbers	N/A

Location of Test: (include cross street if applicable)

Along 108 St. Near Green Bay Rd.

Sketch of Site / Piping Layout

Label: Residual Hydrant, Flow Hydrant Elevations, and Main Sizes



Flow Test Form

Procedure for conducting available water test:

Step 1 Locate personnel at the test hydrant and at all flow hydrants to be used.

Step 2 Remove a hydrant cap from the test hydrant and attach the pressure gauge cap with the petcock in the open position. After checking the second cap for tightness, open the hydrant slowly several turns. Once the air has escaped and a steady stream of water is flowing, the petcock should be closed and the hydrant opened fully.

Step 3 Read and record static pressure as seen on the pressure gauge.

Step 4 The individual at the flow hydrant(s) should remove the cap(s) from the outlet(s) to be flowed. When using a hydrant outlet, the hydrant coefficient and the actual inside diameter of the orifice should be checked and recorded. If a nozzle is placed on the outlet, its' coefficient and diameter should be checked and recorded.

Step 5 Open flow hydrants as necessary, take and record Pitot readings of the velocity pressures. *(Note: Pitot reading should not drop below 10 PSI.)* The individual at the test hydrant should simultaneously read and record the residual pressure. *(The residual pressure should drop at least 15%.)* If it does not drop 15%, open another outlet on the hydrant. *(Also, the residual pressure should not drop below 20 PSI during the test.)* If this happens, reduce the number of flowing outlets. Pitot gauge readings should be between 10 and 30 PSI.

Step 6 Slowly close the flow hydrants, one at a time, to prevent water hammer in the mains. After checking for proper drainage, replace and secure all hydrant caps. Report any hydrant defects.

Step 7 Check the test hydrant for the return to the normal operating pressure, and then close it. The petcock valve should be opened to prevent a vacuum on the gauge. Remove the gauge. After checking for proper drainage, replace and secure the hydrant cap. Report any defects

Note: When flowing water from the large pumper outlet, a second coefficient must be applied in addition to the hydrant butt coefficient.

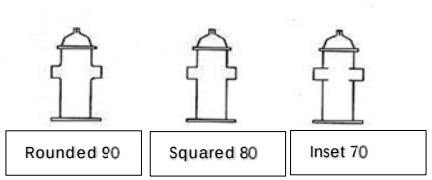
Note: Best results are obtained when the Pitot pressure is between 5 & 10 PSI. Refer to chart below for pumper outlet coefficients:

Pumper Outlet Coefficients

Pitot Pressure PSI	Coefficient
2	0.97
3	0.92
4	0.89
5	0.86
6	0.84
7 and over	0.83

Hydrant Butt Coefficients

Hydrant Butt Coefficients



Note: The 1st hydrant above is Rounded .90
The 2nd hydrant above is Squared .80, and the 3rd hydrant above is inset .70

Note:
The following hydrants that are circled are where the gauge is attached. In general, when flowing a single hydrant, the gauge location hydrant should be between the flow hydrant and the water source.

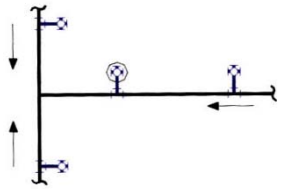
Dead end Main—Water one way



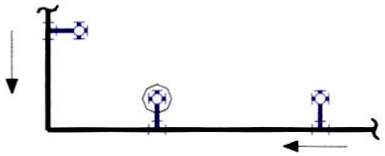
Water from two directions



Water from three directions



Water from two directions at corners

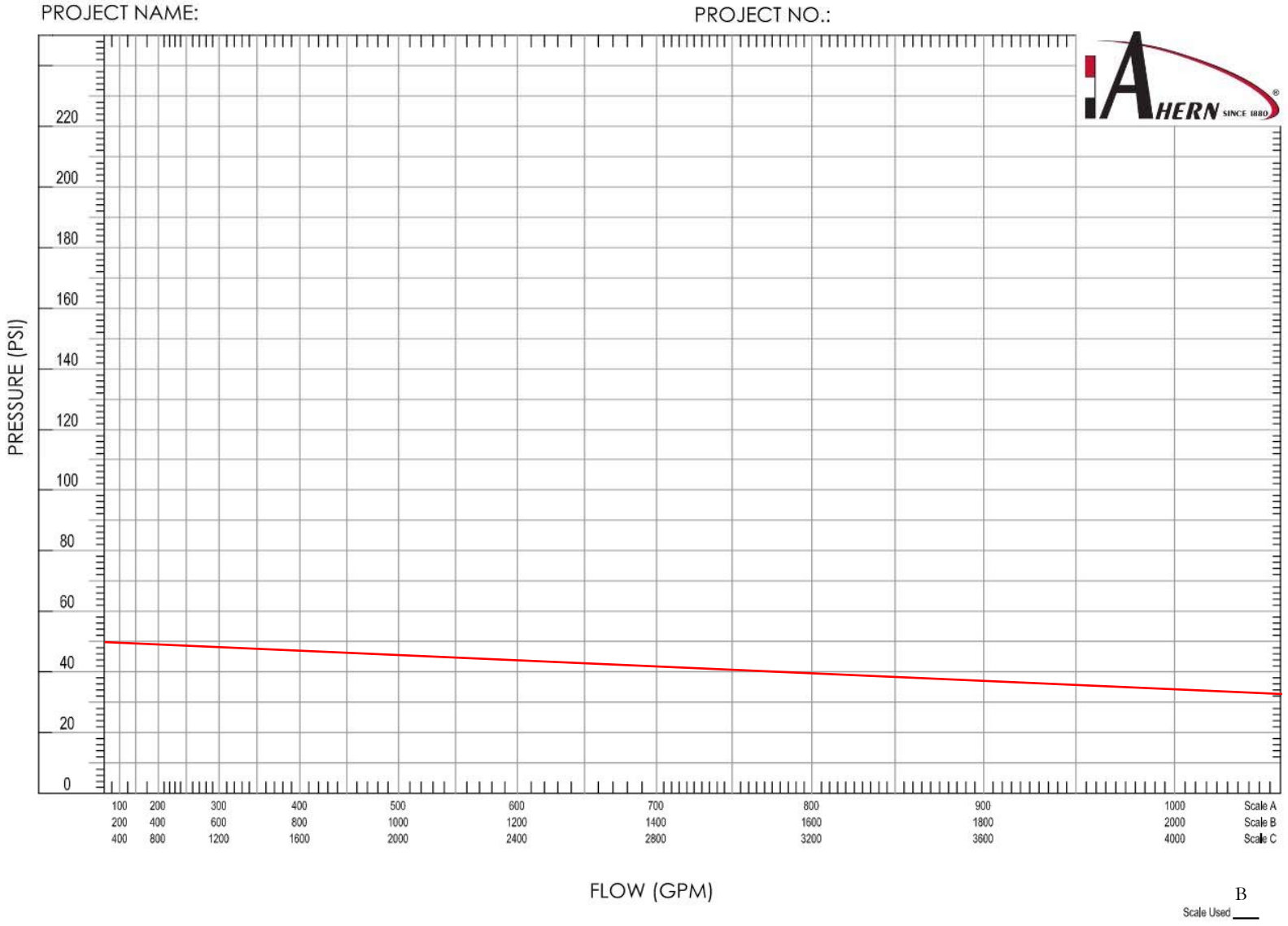


Water from four directions

Form Revision: January 29, 2024

Flow Test Form

PROJECT NAME: PROJECT NO.:



B

Scale Used ___

Form Revision: January 29, 2024

FIRE FLOW TEST WATER PRESSURE LOSS

FOR

Zilber Industrial 3 @ Lakeview Corporate Park

PINNACLE ENGINEERING GROUP
15850 W. Bluemound Road | Suite 310 | Brookfield, WI 53005

DESIGN BY: AJS
PROJECT NUMBER: 644.00
DATE: 10/9/2026

Existing Flow Data

Location of Existing Test	Intersection of 108th St and STH 31, northwest Corner.	
Existing Static Pressure	50.0	PSI
Existing Residual Pressure	41.0	PSI
Existing Flow	2190	GPM
Existing Nozzle Elevation	724.0	

Proposed Flow Data

Location of Critical Hydrant	Flowing Test Hydrant	
Proposed Flow	500	GPM
Proposed Nozzle Elevation	714.9	

Computed Losses/Gains

Residual Pressure Loss				0.6	psi
Fricting Losses in Pipes					
Segment Description	16" in 108th St.				
Diameter	16.00				
Length	445.00				
C Factor	140				
	Segment Loss	0.1	head (ft)	0.03	psi
Segment Description	6" though Hydrant Assemblies				
Diameter	6.00				
Length	33.00				
C Factor	140				
	Segment Loss	0.6	head (ft)	0.26	psi
	Total Pipe Losses			0.3	psi
Static Pressure Loss/Gain Due to Elevation (positive is a loss, negative is a gain)				-4.0	psi
Residual Pressue at Critical Hydrant				**53.1**	psi
Flow at Critical Hydrant				**500**	gpm

C Factors	
Cement Lined Iron	120
PVC	140

WATER PRESSURE LOSS

FOR

Zilber Industrial 3 @ Lakeview Corporate Park

PINNACLE ENGINEERING GROUP
15850 W. Bluemound Road | Suite 310 | Brookfield, WI 53005

DESIGN BY: AJS
PROJECT NUMBER: 644.00
DATE: 10/9/2026

Existing Flow Data

Location of Existing Test	Intersection of 108th St and STH 31, northwest Corner.	
Existing Static Pressure	50.0	PSI
Existing Residual Pressure	41.0	PSI
Existing Flow	2190	GPM
Existing Nozzle Elevation	724.0	

Proposed Flow Data

Location of Critical Hydrant	Southwest corner of the property	
Proposed Flow	500	GPM
Proposed Nozzle Elevation	730.9	

Computed Losses/Gains

Residual Pressure Loss			0.6	psi
Fricting Losses in Pipes				
Segment Description	16" Along STH 31			
Diameter	16.00			
Length	1097.00			
C Factor	140			
Segment Loss	0.2	head (ft)	0.07	psi
Segment Description	6" though Hydrant Assemblies			
Diameter	6.00			
Length	25.00			
C Factor	140			
Segment Loss	0.4	head (ft)	0.19	psi
Total Pipe Losses			0.3	psi
Static Pressure Loss/Gain Due to Elevation (positive is a loss, negative is a gain)			3.0	psi
Residual Pressue at Critical Hydrant			**46.2**	psi
Flow at Critical Hydrant			**500**	gpm

C Factors	
Cement Lined Iron	120
PVC	140